

2.1 Introduction to Vector-Valued Functions_Guide

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2.1
Introducti...

2.1 Introduction of Vector-Valued Functions

Definition: A vector-valued function or vector function, is simply a function whose domain is a subset of the real numbers and whose range is a set of vectors. If the vector function $\mathbf{r}(t)$ maps real numbers to n -dimensional vectors, we write: $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$.

Note: In this class we are most interested in vector functions of the form $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^3$.

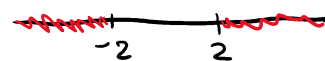
Note: Very strictly speaking, it is a slight abuse to write $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$ when the domain of $\mathbf{r}$ is not the full set of real numbers. In particular if D is the domain of $\mathbf{r}$ and it is a subset of the real numbers but not equal to the real numbers, we should write $\mathbf{r}: D \rightarrow \mathbb{R}^n$. We will ignore this subtlety in this class and be aware that $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$ only means that the domain of $\mathbf{r}$ is a subset of real numbers.

Definition: If a vector function is of the form $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, then $f(t)$, $g(t)$, and $h(t)$ are called the component functions of $\mathbf{r}$.

Note: In this case we have $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^3$ because the real number t gets mapped to a vector $\langle f(t), g(t), h(t) \rangle$.

Example 1: Find the domain of the vector function: $\mathbf{r}(t) = \langle \sqrt{4-t^2}, \cos(t), \sqrt{t^2-1} \rangle$.

Domain for $\sqrt{4-t^2}$ is $[-2, 2]$



Domain for $\cos(t)$ is $(-\infty, \infty)$



Domain for $\sqrt{t^2-1}$ is $(-\infty, -1] \cup [1, \infty)$



$\mathbf{r}(t)$ has domain $[-2, -1] \cup [1, 2]$.

Note: It is important to see that as we graph a vector-valued function in this class we are simply graphing a parametric function in 3-space. Therefore, t will act as a parameter and each vector-valued function will have an associated orientation.

Example 2: We have seen that the parametric equations for the line that passes through the points (2,-3,4) and (4,2,1) are: $x=2+2t$ $y=-3+5t$ $z=4-3t$. How could one represent this line as a vector function?

$$\vec{r}(t) = \langle 2+2t, -3+5t, 4-3t \rangle$$

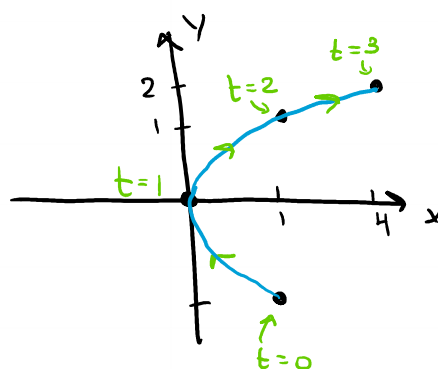
Example 3: Sketch a graph of each of the following functions.

a) $\mathbf{r}(t) = (t^2 - 2t + 1)\mathbf{i} + (t-1)\mathbf{j}$ for $0 \leq t \leq 3$.

$$\text{Note } x = t^2 - 2t + 1 = (t-1)^2$$

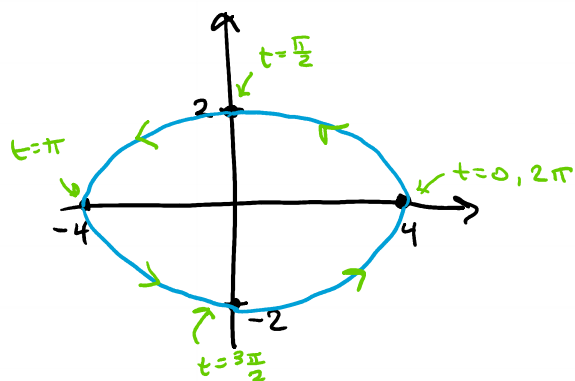
$$\& y = t - 1$$

$$\text{So } x = y^2$$

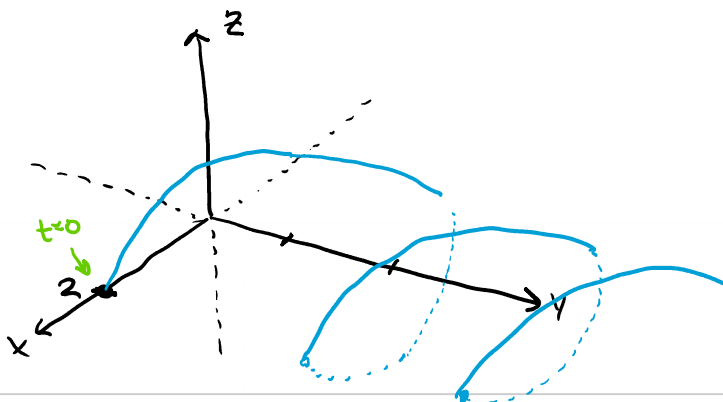


b) $\mathbf{r}(t) = 4\cos(t)\mathbf{i} + 2\sin(t)\mathbf{j}$ for $0 \leq t \leq 2\pi$.

$$x = 4\cos(t) \& y = 2\sin(t)$$



c) $\mathbf{r}(t) = 2\cos(t)\mathbf{i} + 2t\mathbf{j} + \sin(t)\mathbf{k}$ for $t \geq 0$.

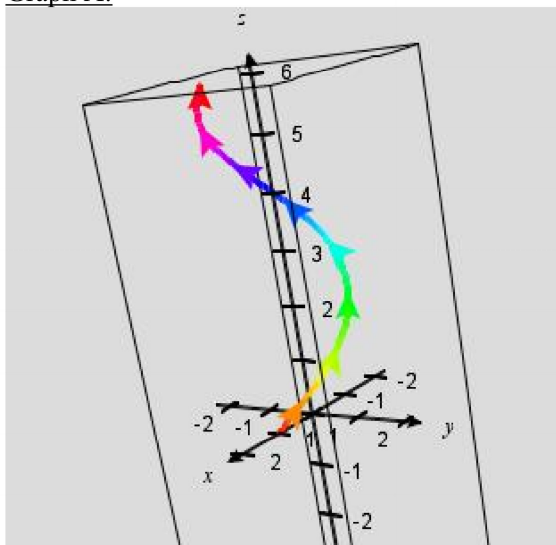


Example 4: Use a computer to graph the vector function:

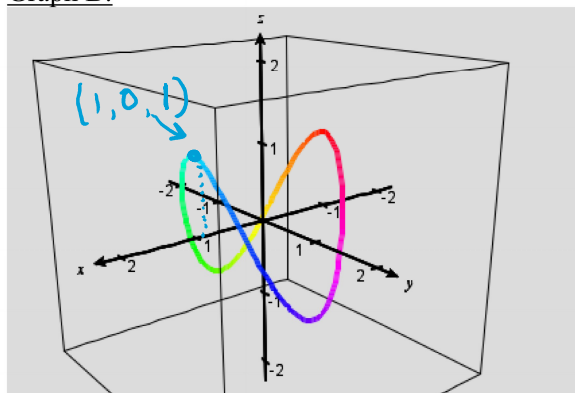
$$\mathbf{r}(t) = \langle (4 + \sin(20t))\cos(t), (4 + \sin(20t))\sin(t), \cos(20t) \rangle$$

Example 5: Match each picture below with its appropriate vector function (you are not allowed to use a computer).

Graph A:



Graph B:



Possible functions:

1. $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle \rightarrow \textcircled{A}$
2. $\mathbf{r}(t) = \langle \cos(t), \sin(t), \cos(2t) \rangle$ if $t=0$ $\vec{r}(0) = \langle 1, 0, 1 \rangle$
3. ~~$\mathbf{r}(t) = \langle \cos(t), t, \sin(t) \rangle$~~ $\rightarrow \textcircled{B}$
4. ~~$\mathbf{r}(t) = \langle \cos(t), \cos(t), \cos(t) \rangle$~~ if $t=0$ $\vec{r}(0) = \langle 1, 1, 1 \rangle$